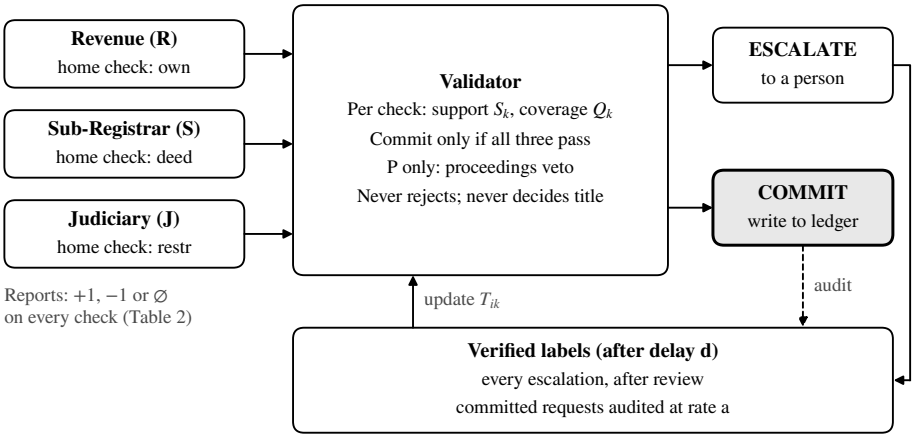


Fig. 1. Validation loop. Three offices report on three checks, the validator commits a request only when all three checks pass and otherwise escalates it, and verified labels update trust.

LNCS fit check

Each float below is included exactly as in the paper. An overfull box warning in the log means a float is wider than the 122 mm text area.

Table 1. Comparison with related work

Work	Land records	Multi-source	Per-claim authority	Learned trust	Pre-commit conflict check	Escalates	Ground truth
Thakur et al. [1]	yes	yes (nodes)	no	no	no (deferred)	manual	no
Soner et al. [5]†	yes	yes	?	?	?	?	?
Shuaib et al. [6]†	yes	?	?	?	?	?	?
Sahai, Pandey [7]†	yes	?	?	?	?	?	?
Khanna et al. [15]‡	yes	yes (6 roles)	role-based	n.s.	n.s.	n.s.	n.s.
Chainlink [10]	no	yes (nodes)	no	reput.	aggreg.	no	no
Truth discovery [13]	no	yes	no	yes (unsup.)	yes	no	—
DecenTruth [14]‡	no	yes	no	yes (unsup.)	yes	n.s.	sim.
This work	synthetic	yes (3)	yes	yes (labels)	yes	yes	yes

† Full text not yet read; ? = not determined. ‡ Compared at abstract level only. n.s. = not stated in the abstract; unsup. = estimated without labels; reput. = reputation; aggreg. = aggregation of node reports; sim. = simulation; — = not applicable to a survey.

Table 2. Illustrative meaning of each simulated report ($+1$ / -1; $\varnothing$ = no response)

Office	own: seller is recorded holder	deed: deed registered, matching	restr: no pending proceeding
Revenue	Home. $+1$ record of rights names the seller; -1 names another	$+1$ mutation file cites a matching deed; -1 no match	$+1$ no pending revenue proceeding noted; -1 one is noted
Sub-Registrar	$+1$ earlier registered documents consistent with seller; -1 not	Home. $+1$ deed registered, parties and parcel match; -1 not	$+1$ no notice of a pending suit noted; -1 one is noted
Judiciary	$+1$ no court record contradicts the holding; -1 a decree names another holder	$+1$ no case challenges the deed; -1 one does	Home. $+1$ nothing pending in any forum covered; -1 a proceeding is pending

Simulated signals only; not claims about the records real offices hold or certify.

Table 3. Claim-specific authority $A(i, k)$ (modelling assumption)

Check	Revenue	Sub-Registrar	Judiciary
own	1.0	0.6	0.4
deed	0.6	1.0	0.4
restr	0.2	0.2	1.0

Table 4. Worked example: request 123 in S5a, seed 1, with a pending proceeding visible only to the Judiciary

Mechanism	Support for 'clear' on restr	Decision	Reason code	Outcome
B1 Majority	2 of 3 clear	COMMIT	—	false commit
B2 Authority-only	0.286	ESCALATE	restr: low support	escalated
B3 Trust-only	0.666	COMMIT	—	false commit
P−V Hybrid	0.285	ESCALATE	restr: low support	escalated
P Hybrid + veto	veto first	ESCALATE	veto: proceeding reported	escalated

Truth: own pass, deed pass, restr pending (request invalid). Reports (own, deed, restr): R: +1 +1 +1; S: −1 +1 +1; J: +1 +1 −1. Trust under ideal feedback at that point: R, S and J on restr $\approx$ 1.00; $\theta = 0.65$.

Table 5. Scenarios; each changes one setting from S1

Scenario	Change from S1	RQ
S1 Aligned	None: authority matches each office's most accurate check	1
S2 Degraded	From request 251, Revenue accuracy 0.50 on all checks	2
S3 Misaligned	On own, Revenue 0.75 and Sub-Registrar 0.95; authority unchanged	2
S4 Low availability	Outage probability 0.25 per office per request	3
S5a Proceedings stress	Revenue and Sub-Registrar always report 'clear' on restr	4
S5b Alt. authority	As S5a, with A(R, restr) = A(S, restr) = 0.6	4

Table 6. Mechanisms compared

Mechanism	Weight	Pass rule per check	Coverage gate	Veto	Learns
B1 Majority	1	$>$ half of responders	$\geq$ 2 of 3 responded	no	no
B2 Authority-only	$A(i,k)$	$S_k \geq \theta$	$Q_k \geq 0.5$	no	no
B3 Trust-only	T_{ik}	$S_k \geq \theta$	$Q_k \geq 0.5$	no	yes
P−V Hybrid	$A(i,k) \cdot T_{ik}$	$S_k \geq \theta$	$Q_k \geq 0.5$	no	yes
P Hybrid + veto	$A(i,k) \cdot T_{ik}$	$S_k \geq \theta$	$Q_k \geq 0.5$	yes	yes

Table 7. False-commit rate (FCR) and valid-commit coverage (VCC), %, mean over 100 seeds, ideal feedback

	S1 Aligned		S2 Degraded		S3 Misaligned		S4 Low avail.	
Mechanism	FCR	VCC	FCR	VCC	FCR	VCC	FCR	VCC
B1 Majority	3.05	73.2	6.34	56.3	3.03	73.2	2.87	44.6
B2 Authority-only	1.15	71.5	4.63	44.8	2.09	60.0	1.64	45.9
B3 Trust-only	1.06	61.7	2.54	42.2	1.03	61.7	1.35	39.5
Hybrid (P−V = P)	1.09	68.7	3.86	44.2	1.50	57.9	1.13	31.8

Between-seed SD: FCR 0.45–1.23, VCC 1.7–2.0. P and P−V made identical decisions in every run of S1–S4.

Table 8. Pre-specified paired comparisons (100 seeds; percentage points; breaches per 1,000 requests)

ID	Metric: comparison (scenario)	Difference [95% CI]	Half-width	Holm p	Reading
H1a	FCR: P−V − B1 (S1)	−1.96 [−2.12, −1.81]	0.16	<0.001	lower
H1b	FCR: P−V − B2 (S1)	−0.06 [−0.09, −0.03]	0.03	<0.001	lower, $< \Delta$
H2a	FCR: P−V − B2 (S2, 251–1000)	−1.28 [−1.49, −1.08]	0.20	<0.001	lower
H2b	VCC: P−V − B3 (S2, 251–1000)	+0.30 [−0.13, +0.72]	0.42†	0.25	no clear diff.
H2c	FCR: P−V − B2 (S3)	−0.60 [−0.67, −0.52]	0.08	<0.001	lower
H3	VCC: P−V − B1 (S4)	−12.82 [−13.10, −12.54]	0.28†	—	lower (descr.)
H4	Breaches: P − P−V (S5b)	0	0.00	1.00	identical
H5	FCR: P under F1 − F0 (S2)	−0.13 [−0.25, −0.005]	0.12	1.00	not supported

† Above the precision target $\Delta/2$ at the 100-seed cap. H3 is descriptive. H5 was one-sided (F1 higher).

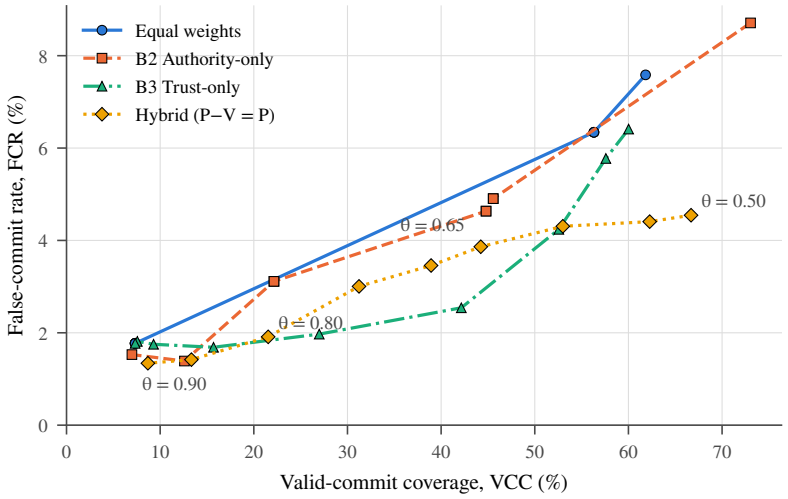


Fig. 2. Commit-threshold trade-off after Revenue degrades (S2): false-commit rate against valid-commit coverage as θ varies from 0.50 to 0.90 in steps of 0.05 (means over 100 seeds, ideal feedback)

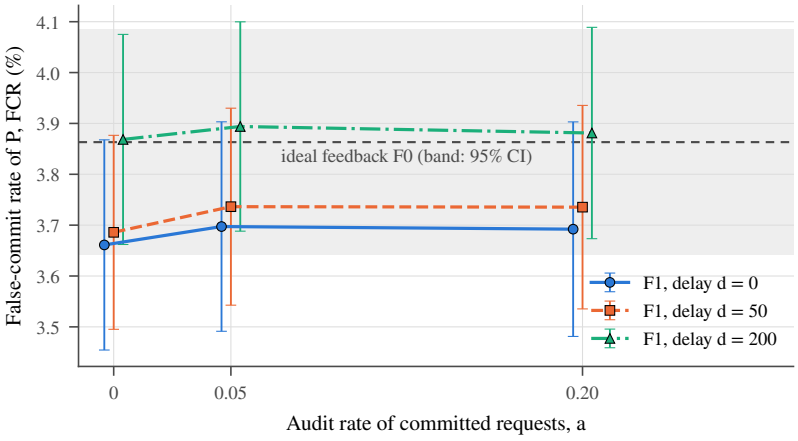


Fig. 3. False-commit rate of P under realistic feedback F1 in S2, by audit rate and label delay; whiskers are between-seed 95% CIs, the dashed line and band show ideal feedback F0

Table 9. Exploratory coverage-gate ablation A1 (paired 95% CIs, 100 seeds, percentage points)

Comparison	Scenario	FCR difference	VCC difference
Hybrid − majority, no gate (G0)	S1	−1.85 [−2.00, −1.69]	−0.27 [−0.53, −0.004]
Hybrid − majority, no gate (G0)	S4	−0.99 [−1.14, −0.83]	+8.41 [+8.13, +8.70]
Hybrid − majority, common count gate (G1)	S1	−1.85 [−2.01, −1.70]	−0.27 [−0.53, −0.004]
Hybrid − majority, common count gate (G1)	S4	−1.01 [−1.19, −0.83]	+8.41 [+8.13, +8.70]
Hybrid: weighted gate − count gate	S1	−0.11 [−0.15, −0.07]	−4.25 [−4.39, −4.11]
Hybrid: weighted gate − count gate	S4	−0.72 [−0.85, −0.60]	−21.23 [−21.52, −20.95]

Table 10. Exploratory robustness study A2: levels at which each main conclusion holds, with the range of FCR differences in points

Factor varied	C1 (S1)	C1 (S2)	C2	C3
Off-home accuracy 0.60–0.85	4/4 (−6.2 to −0.4)	4/4 (−4.9 to −1.4)	3/4 (−1.6 to −0.2)	4/4 (−1.3 to −0.8)
Home accuracy 0.85–0.95	3/3 (−2.0 to −0.9)	3/3 (−2.5 to −1.6)	3/3 (−1.3 to −1.1)	3/3 (−1.9 to −1.3)
Shared R+S errors ρ = 0–0.20	4/4 (−3.6 to −2.0)	4/4 (−3.6 to −2.5)	4/4 (−2.0 to −1.3)	4/4 (−5.0 to −1.3)
Label noise e = 0–0.10, F0	—	4/4 (−2.5 to −2.4)	4/4 (−1.3 to −1.1)	4/4 (−1.3 to −1.2)
Label noise e = 0–0.10, F1	—	4/4 (−2.6 to −2.5)	4/4 (−1.6 to −1.4)	4/4 (−0.9 to −0.8)

C1: hybrid below majority. C2: hybrid below authority-only after degradation (requests 251–1000). C3: trust-only below hybrid (S2). Holds: the 95% CI excludes 0 with the main-study sign. No conclusion reversed.

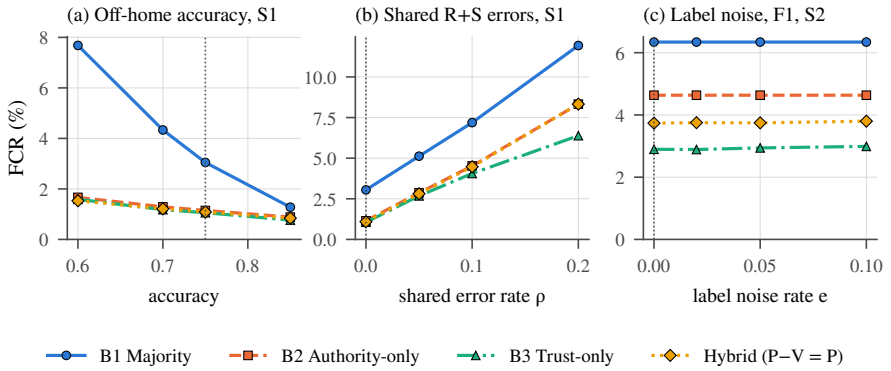


Fig. 4. Exploratory robustness study A2 (100 seeds): false-commit rate by mechanism as (a) off-home accuracy, (b) the shared error rate of Revenue and Sub-Registrar and (c) label noise vary; dotted lines mark the main-study settings

Table 11. Status of every result

Result	Status
Tables 7–8; Figs. 2–3; $q_{\min}$ sweep	Pre-specified (frozen protocol); five analysis choices in Table 12
Veto bound (Sect. 3)	Analytic, derived before the protocol freeze
Counts per 1,000 requests; quoted trust values	Descriptive re-expression of pre-specified runs
Two-sided test for H5	Exploratory, added after results were seen
Table 9 (A1); Table 10 and Fig. 4 (A2)	Exploratory; addenda committed before running

Table 12. Analysis choices not covered by the frozen protocol (accepted by the supervisor)

#	Situation	Choice
1	H4 has zero variance	Interval is the point 0; p = 1.0 in the Holm family
2	H1b has no direction	Two-sided test inside the Holm family of seven
3	Significant and inside $\pm\Delta$ (H1b)	Worded 'lower, but smaller than Δ'
4	CI opposite a one-sided hypothesis (H5)	'Not supported'; two-sided CI and exploratory test reported
5	H3 has no direction	Worded from its unadjusted CI; labelled descriptive